

Highlights

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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- Scoring the same model at the fire, not region-wide, costs 0.133 AUC
- The cause is the negative pool, not prevalence: +0.149 against -0.002
- Equalising the evaluation frame lifts mean transfer from 0.527 to 0.589
- Across 20 transfer directions the thermal gain is +0.007 [-0.021, +0.037]
- Highest niche-overlap pair fails both ways; lowest-overlap pair transfers both ways

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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Abstract

We test whether six pre-fire thermal predictors, added to a terrain, fuel and greenness baseline, transfer between five Mediterranean regions under MCD64A1 labels and spatially blocked validation. The first result is **evaluation geometry**: with the model held fixed, scoring on the burn scar and its 2 km collar rather than region-wide costs **0.133 ROC-AUC** [**+0.059, +0.207**]. Equalising the study areas to a 10 km collar lifts mean transfer from 0.527 to 0.589. Local skill does not travel: the thermal block adds +0.045 to +0.148 within regions at 5 km blocking but +0.007 [−0.021, +0.037] across twenty directions, and the static baseline transfers no better, at 0.519 against 0.527. At the point estimates, similarity is neither sufficient nor necessary: the most niche-similar pair fails both ways (0.438, 0.345), the least similar transfers above chance (0.594, 0.548). Transfer skill must be measured on the target region before a model is relied on.

Keywords: wildfire, model transferability, model evaluation, land surface temperature, domain adaptation, concept shift

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1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns (Pausas and Keeley, 2021). The dominant pattern in fire susceptibility mapping (Vibhandik et al., 2026; Jodhani et al., 2026) is settled. Geospatial predictors are assembled over a study region, paired with a historical burned-area record and fitted with a supervised classifier, most often a random forest (Breiman, 2001). The surface is then published with a cross-validated AUC of 0.85 to 0.95. **That pattern does not report portability**, and this paper measures it.

10 1.1. *Portability goes unmeasured*

A susceptibility model’s reported skill is almost always a *within-region* estimate: held-out folds come from the same study area and season, often the same fire, and random folds over cells let spatial autocorrelation inflate it further. The problem is documented across ecological modelling (Roberts et al., 2017; Ploton et al., 2020) and addressed by spatially blocked cross-validation (Valavi et al., 2019; Meyer et al., 2018), but blocking does not make the estimate honest about a fire the model has not seen.

A predictor block is adopted on the increment it delivers inside its training footprint; whether that survives a change of region is not asked. Yet any regional product built from locally trained models implicitly promises generalisation beyond it. Meteorological fire-danger indices are known not to port cleanly in a Peruvian case study (Podschwit et al., 2022). The two studies that test transfer systematically (Dimarco et al., 2026; Liu and Mostafavi, 2025) both report that it largely succeeds between similar regions. Both transfer models whose dominant predictors are *spatially stationary*. Whether a dynamic, season-specific class behaves the same way is this paper’s ques-

tion.

1.2. Pre-fire thermal dryness is the natural test case

Pre-fire thermal dryness is the dynamic class most plausibly *expected* to
30 transfer. Standard predictors are static or near-static over the timescale at
which fire danger varies, so they explain poorly why one summer burned and
the preceding one did not. What changes is the state of the surface, to which
satellite thermal observation gives partial access (Section 2.2). The physics
linking moisture stress to combustion is universal, which makes the class
35 diagnostic: a loss cannot be dismissed as a peculiarity of a locally defined
covariate. The expectation is sharpest for the internally normalised chan-
nels, which should be least exposed to absolute-temperature offsets between
regions; **they transfer no better than the absolute ones** (Section S1.6).
The expectation motivates the design and is not a finding of it. Section 4.4
40 reports the associations running the other way.

1.3. Contributions

Three findings carry this paper, stated here as claims and established in
Section 4, which carries every interval.

Contribution 1. Where a model is scored decides what it ap-
45 **pears to know, and the effect is large enough to dissolve findings**
of our own. That evaluation extent inflates AUC is established in species
distribution modelling (Section 2.4). What is new is a magnitude under a
controlled design. Holding model, predictors and fitting fixed and changing
only which cells are scored costs **0.133 ROC-AUC**, the size of the incre-
50 ments this design measures for a predictor block. The cause is measured as
the composition of the negative pool rather than class balance (Section 4.3),
and the cost holds with the region as the resampling unit. Applied between

regions, the same test shows four of our own quantities to be properties of the frames, including the one arm that held place fixed (Section 4.4). One reversal survives it, Manavgat’s elevation. The practical consequence is a reporting standard (Section 5.6).

Contribution 2. Local skill does not travel, and correcting the frame does not rescue it. The thermal block is worth a substantial within-region increment in all five regions, every bootstrap interval above zero, and it stays positive when the frame is equalised. At the scar level much of it is a property of interleaved holdout; clustered by region, that part is not established (Section 4.3). Across twenty transfer directions the paired contribution spans zero on both frames under the primary resampling unit. Once equalised it lies much closer to the boundary, and under one admissible unit, clustering by target region, it excludes zero (Section 4.4). Its sign belongs to the pair rather than the block. Two controls bound this. The static baseline transfers no better on either frame, so the failure is not the thermal block’s peculiarity. On matched frames and blocking, equalised transfer still falls **0.197** short of the within-region reference. The within-region half is not novel (Alkan Akıncı and Akıncı, 2023; Iban and Sekertekin, 2022); the paired contrast against portability is.

Contribution 3. No similarity diagnostic we tested is shown to order transfer, conditional ones included. Twenty measures were fixed in advance. They range from marginal covariate distance through niche overlap and regime structure to the conditional agreement of predictor–burning relationships. Under the corrected label nineteen have correlation intervals spanning zero. The twentieth is defined in only six directions, with a degenerate interval, and is not interpreted (Section 4.6, Section S4). The conditional measure that had ordered transfer under the frozen label ($\rho = +0.84$)

no longer does $(+0.52 [-0.27, +0.87])$. With ten effective region pairs this is a failure to show ordering, not proof that none exists. The practical consequence is that no pre-deployment shortcut replaces measuring transfer on the target.

Two consequences follow, as supporting results. Label-free alignment by standardisation and covariance alignment (Sun et al., 2016) does not repair transfer, in what we believe is its first application to fire susceptibility. And because the residual is conditional, the resource that closes it is target labels, priced in Section S1.19.

The leakage-audited, spatially blocked protocol is released with code, configuration and frozen outputs, so most of this can be re-run rather than taken on trust. The release is complete, as the declarations record. That matters given evidence that wildfire transfer conclusions are sensitive to evaluation design (Xu et al., 2026). A companion paper treats the observational layer beneath this one.

2. Related work

2.1. Fire susceptibility mapping, and what it reports

The dominant pattern is described in Section 1; the figure it publishes, a cross-validated AUC of 0.85 to 0.95, is a within-region estimate. Comparable within-region results already exist for the landscapes studied here (Alkan Akıncı and Akıncı, 2023; Iban and Sekertekin, 2022).

2.2. Pre-fire thermal dryness

Satellite thermal observation gives repeated access to surface state through fuel moisture content (Yebra et al., 2013). Chuvieco et al. (2004) established the pairing of land surface temperature with a vegetation index

105 as a live-fuel-moisture estimator for fire-danger rating. The Temperature-Vegetation Dryness Index (Sandholt et al., 2002) formalises that feature space into an internally normalised measure, which is why it is often expected to travel better than raw temperature. That pre-fire thermal state carries information about subsequent fire is established (Maffei et al., 2018; 110 Maffei and Menenti, 2019; Maffei et al., 2021). Gelabert et al. (2025) found dead fine fuel moisture and its anomalies the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether such a classifier survives strict, label-free application to an unseen region.

115 2.3. *Spatial validation and transferability*

The canonical taxonomy separates covariate shift from concept shift (Moreno-Torres et al., 2012). Under covariate shift the predictor distribution moves but the predictor-response relationship holds; under concept shift the relationship itself changes. Covariate shift is in principle correctable without 120 target labels, by per-region standardisation or covariance alignment such as CORAL (Sun et al., 2016), and domain adaptation has an established remote-sensing literature (Tuia et al., 2016; Persello and Bruzzone, 2012). Concept shift is not: no realignment of inputs can repair a reversal in the sign of an association, because detecting one requires the labels being 125 withheld. Shift decomposition in applied remote sensing is not itself new (Huang et al., 2026), but we are aware of no prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction.

2.4. *Cross-region generalisation of fire models*

Few studies test fire-model transfer directly. Podschwit et al. (2022) 130 report, in a Peruvian case study, that meteorologically derived danger in-

dices do not port cleanly between fire environments. WildfireGenome (Liu
 and Mostafavi, 2025) runs a leave-one-county-out matrix across seven US
 counties, and reports strong within-county performance with highly variable
 off-diagonal transfer, similar pairs transferring well and dissimilar pairs col-
 lapsing. Its label is a principal-component composite of hazard *indicators*
 rather than observed burned area, and it applies no adaptation. Xu et al.
 (2026) argue that wildfire transfer conclusions depend strongly on evaluation
 design. We address that caution in two ways: the protocol was fixed in a
 project log before the diagnostics were computed, and every sensitivity axis
 is reported. That log is not a formal pre-registration, and Section S4 states
 what was fixed and when. Kondylatos et al. (2023) provide Mesogeos, a 1
 km Mediterranean datacube.

Evaluation extent and AUC. Species distribution modelling settled
 long ago that the area a model is evaluated over is not a neutral choice.
 Lobo et al. (2008) make it the fifth and, by their own ranking, most im-
 portant reason to distrust AUC comparatively. The extent of the modelled
 area governs how many easy absences enter the calculation, and therefore the
 score. VanDerWal et al. (2009) show the same lever on the calibration side,
 and Barve et al. (2011) give the argument its general form as the accessible
 area. That literature is qualitative about magnitude, because magnitude is
 problem-specific, so our Contribution 1 is a measurement inside that frame-
 work rather than a new phenomenon. The wildfire literature has largely
 not imported the lesson: region-wide figures are reported as though they
 described performance at the fire. We know of no wildfire study that holds
 the model fixed and varies only the evaluation cells. That is why Section 5.6
 treats the 0.133 as a reporting problem rather than a caveat.

The nearest neighbour, and the contrast this paper draws. Di-

marco et al. (2026) is the closest Mediterranean analogue. They harmonise 500 m predictors across four countries and fit tree ensembles under spatial cross-validation. Transfer is tested both leave-one-country-out and as a full 4×4 matrix. Every transfer exceeds AUC 0.80, bioclimatically similar countries score higher, and no domain adaptation is used.¹ Much of the design is shared: Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation and an explicit transfer matrix; two things differ. **The predictor class:** their model rests on attributes of a place, all spatially stationary, ours on the state of a surface in one season. **The response variable:** theirs is human-driven ignition, ours burned area, which have different dominant controls. Read together, the two results suggest the relationship between domain similarity and transfer success may be predictor-class dependent. Vesk et al. (2021) align with that reading from species distribution modelling, while Dimarco et al. and WildfireGenome run against it. Rousseau and Betts (2022) found environmental similarity not a significant predictor of transferability; our result is consistent with theirs. We present that as a live disagreement; Section 5.5 states why our data do not settle it.

3. Data and methods

3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed (Fig. 1): Manavgat 2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangle

¹Their Results text refers to “LST anomalies”, inconsistent with their own Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology (Muñoz-Sabater et al., 2021). We follow their Methods and note the discrepancy so the comparison is transparent.

in EPSG:4326, defined from place coverage rather than from a fire perimeter and deliberately not clipped to it, so that unburned cells around each fire form the negative class. **The record does not show every box fixed before any outcome was seen**, so what it does show is stated region by region. In the pipeline’s version history each box has a single committed value, never changed afterwards. Montiferru’s is derived deterministically from the union of four municipal boundaries. Manavgat’s was drawn to exclude the coastal cropland belt; its final coordinates are in a dated AOI preview record of 7 July 2026, the day before its first gate result. Bejís’s is still labelled the initial candidate in the registry but was committed after its first gate and model results. Muğla’s coordinates appear in a dated preflight record about four minutes before its first gate result and were committed to the registry only afterwards. No box needed adjusting to pass the gate, whose admitted margins are wide (Section 4.1). **One choice was label-informed and is stated as such:** the North Evia box was extended after the legacy box proved atypically high in burned prevalence, the extended geometry then defined from place anchors and the legacy variant kept as a sensitivity arm. Its consequences are in Section 4.4 and Section S3.5(ix). A sixth region, Kozan 2023, is carried as a negative control and excluded by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Lengths vary with the event: 57 to 61 days for predictors, 35 to 59 for labels, counted inclusively. A four-year baseline of window-symmetric composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and reproduced in Section S3, Table S19. The processing

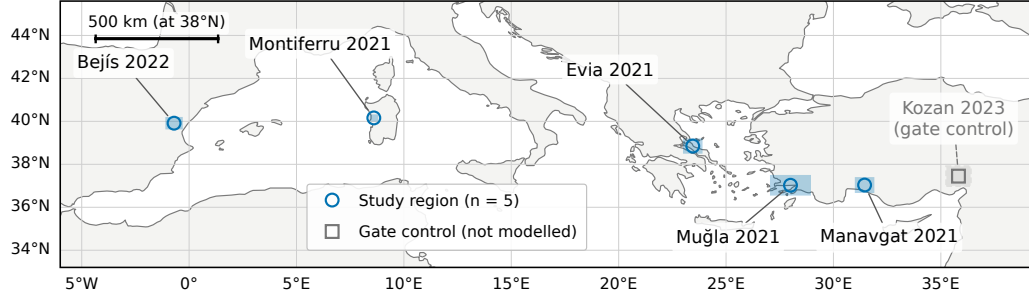


Figure 1: **Study regions and the gate negative control.** Locations of the five wildfire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejis 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table S19 to 5×10^{-3} degrees.

chain and the evaluation programme it feeds are drawn in Fig. 2.

3.2. Burned-area label and the ~ 500 m analysis grid

210 Labels come from MODIS MCD64A1 Collection 6.1 (Giglio et al., 2018) retrieved through Google Earth Engine (Gorelick et al., 2017), whose omission and commission characteristics (Boschetti et al., 2019) bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks,
 215 giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the label window; the label never affects eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning before the label window

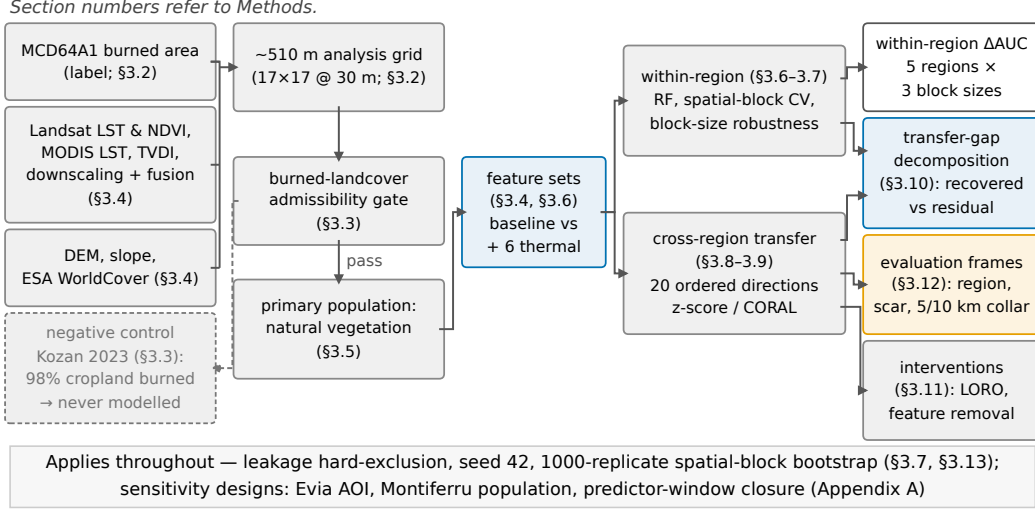


Figure 2: **Processing pipeline and evaluation programme.** Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.10). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (Section S1). Section references, in the figure and in this caption, are verified against the Methods heading list when the figure is built.

220 opens are removed, and burning in earlier years is screened for; the counts by region are in Section S3.1.1. Section S3.1 gives the full specification, including what follows from the grid’s shape.

The Manavgat 2021 label is a corrected one. The label first used for Manavgat was a stale export that missed the fire’s first four days, 28
225 to 31 July. The corrected label only adds burned cells, raising the primary population’s burned count from 784 to 2,935; the correction, the re-freeze and its control arm are specified in Section S3.1.2.

3.3. Burned-landcover admissibility gate

Before any modelling each region passes a gate asking what fraction of
230 its burned cells is dominated by natural vegetation, using ESA WorldCover classes (Zanaga et al., 2022) on the same cells. A region is admitted at 0.50 with at least 30 burned cells, and rejected as a cropland-dominated control at 0.50 cropland. Verdicts and the purpose of the gate are in Section 4.1, the full rule in Section S3.1.

235 3.4. Predictor variables

Two feature sets are used. The **baseline** is terrain, fuel and green-
ness: elevation, slope, dominant land cover and a predictor-window median
vegetation-index composite, all static or near-static over the timescale at
which fire danger varies. The **thermal** set adds six pre-fire channels: current
240 land surface temperature, its anomaly against a four-year window-symmetric
baseline at the same cell, the Temperature-Vegetation Dryness Index and its
difference against that baseline, and two coordinate-informed products, a
downscaled and a fused surface temperature. The two differenced channels
are the ones constructed to isolate the dynamic anomaly. TVDI’s wet and
245 dry edges are percentiles of the values a given area and window happen to

contain, so **it is not portable as a physical quantity independently of any concept shift** (Sections S3.4, S3.5).

Quality screening of the coarse thermal input is compared in Section S1.5, *Quality screening, and a correction*: the two arms ran different
250 step7 versions, and the result does not change (elevation 0.232; no other signed AUC moves by more than 0.005).

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~ 510 m cell from valid 30 m pixels only, with the valid fraction recorded. A cell is `valid_for_modeling` when it
255 is not excluded for pre-label burning and its predictors are valid: joint finite NDVI, elevation and slope support over at least 30 % of the cell, finite means for those three channels, and at least one valid land-cover pixel. **Thermal completeness is not part of the definition**: depending on the region, 6 % to 58 % of valid cells carry at least one missing thermal channel, and
260 missing values are imputed inside the fitting pipeline (Section 3.6) rather than by excluding the cell. The **primary** population is natural vegetation, cells whose combined tree, shrub and grass fraction reaches 0.50, which excludes cropland from every burnable mask. The **secondary** population is all valid cells; the frozen export carries it for the within-region arm in two regions as a
265 sensitivity (Section S1.17); the transfer matrix uses the primary population only.

3.6. Classifier

The thermal set nests the baseline (Section 3.4). Land cover is one-hot encoded. Missing numeric values are median-imputed and the categorical
270 channel most-frequent-imputed, with the imputers fitted inside each training fold, and on the source alone in transfer, so no held-out statistic enters a

fit in the raw arms. Features are not otherwise standardised. The adapted arms of Section 3.9 are the exception, and fill missing values with the region’s own mean. The classifier is a random forest (Breiman, 2001) with 300
 275 trees, unlimited depth, `min_samples_leaf = 3`, balanced class weights and `random_state = 42`, identical in every arm, so that no comparison here is confounded by a model choice.

3.7. *Spatial-block cross-validation and bootstrap uncertainty*

Cross-validation is spatially blocked. Cell i at grid position (r_i, c_i) , from
 280 `row_500m` and `col_500m`, is assigned to the block

$$b_k(i) = (\lfloor r_i/k \rfloor, \lfloor c_i/k \rfloor). \quad (1)$$

Five folds are drawn over whole blocks, stratified by label, so no block is split between training and test. The block size k is reported at 2, 10 and 20 cells, about 1, 5 and 10 km.

Every score is a ROC-AUC on a named set of cells F , its **evaluation**
 285 **frame**. With F^+ the burned and F^- the unburned cells of F , and s_i the predicted score,

$$\text{AUC}(s; F) = \frac{1}{|F^+| |F^-|} \sum_{i \in F^+} \sum_{j \in F^-} \left[\mathbf{1}(s_i > s_j) + \frac{1}{2} \mathbf{1}(s_i = s_j) \right]. \quad (2)$$

This is the probability that a burned cell in F outranks an unburned one, so changing the frame changes the metric even when no score changes (Section 3.12).

290 Uncertainty is a spatial-block bootstrap: the blocks of (1) are resampled with replacement, 1000 replicates, seed 42, and the interval is the 2.5 and 97.5

percentiles. Differences are paired within replicates, and a verdict resting on too few positive-carrying blocks is stated as indicative. Direction-level intervals resample the ten unordered region pairs, and a quantity with one value per scar or target region gets a Student t interval; the mechanics are in
 295 Section S3.6.1. **The effective sample is thus ten pairs or five regions for direction-level intervals, and at most seven scars from three regions for scar-level ones.**

3.8. Cross-region transfer protocol

300 For each ordered pair the model is fitted on the source population and applied to the target with **no target label used at any point**: no refitting, no threshold selection, no calibration. Both feature sets are transferred, so the thermal block’s contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions
 305 are computed.

3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets. **Region-wise z-score** standardises each region’s numeric features with its own statistics, never pooled. **CORAL after region-wise z-score** then
 310 aligns the source covariance to the target’s by the whitening-recolouring map (Sun et al., 2016), with $\lambda = 10^{-5}$, applied to the source only. Target feature statistics, never target labels, enter the adapted arms, and both variants are verified label-blind at run time. λ sensitivity was assessed over nine values on four of the twenty directions, moving transfer AUC by at most
 315 0.014, and no value of λ was selected on performance (Section S1.2). The equations and numerical details are in Section S3.8, *Label-blind adaptation, full specification*.

3.10. Transfer-gap decomposition and the concept-shift criterion

For a direction into a target, let A_w be the target’s within-region thermal
 320 AUC under blocked cross-validation, A_{raw} the raw transfer AUC and A_{ad} the
 transfer AUC after adaptation method m , all on the same target cells. The
 gap is decomposed as

$$G = A_w - A_{\text{raw}}, \quad R_m = A_{\text{ad}} - A_{\text{raw}}, \quad U_m = A_w - A_{\text{ad}}, \quad \rho_m = R_m/G, \quad (3)$$

so $R_m + U_m = G$. The recovered fraction ρ_m is signed and unclipped.
When adaptation lowers AUC, R_m and ρ_m are negative and re-
 325 **ported as negative recovery, never set to zero.** No fraction is reported
 when $G \leq 0$. Intervals come from the paired spatial-block bootstrap of
 Section 3.7 on 2-cell target blocks, resampling within-region out-of-fold and
 transfer scores together and evaluating (3) per replicate (degenerate repli-
 cates: Section S3.7). Where one method is shown per direction it is the one
 330 with the higher A_{ad} , a choice that uses target labels. Section 4.3 shows the
 remainder should not be read as a conditional residual, because much of it
 is incurred inside a single region (Sections S1.10, S3.7).

The mechanism is diagnosed by **signed univariate association**: the
 raw ROC-AUC of each numeric predictor against **burned** is computed per
 335 region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a
 direction rather than weakness. A reversal is called bootstrap-supported only
 when the two regions’ point estimates fall on opposite sides of 0.5 **and each**
region’s own interval excludes 0.5. That is stricter than requiring the
 intervals to be disjoint, and a feature failing the second condition is recorded
 340 as a point reversal only.

3.11. Interventions

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and evaluated on the held-out region, with folds blocked as before. This asks
 345 whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors reversing with bootstrap support **on the frames as drawn** are **elevation_mean** and **lst_anomaly_mean**. Section 4.4 withdraws that support under an equalised frame, so the selection rule is frame-dependent and this arm is
 350 reported as a measurement under the original protocol, not as a consequence of an established reversal. Both are dropped and everything refitted, within-region and across every direction, and each is dropped singly so the cost can be attributed. The two features are selected by the same reversal analysis the result is then read against, so **both quantities are post-selection**
 355 **estimates** with no correction applied.

3.12. Evaluation frames and controls on the transfer path

Let V_R be the primary population of region R and $P_R \subset V_R$ its burned cells. By (2), a frame fixes which burned and unburned cells are compared. The **region-wide frame** is V_R , the rectangle as drawn, used in Sections 4.2
 360 and 4.5.

Scar frames. A scar is an 8-connected component K of P_R with at least 50 cells. Its **scar + 2 km collar** is

$$H_K = \left\{ i \in V_R : \min_{j \in K} (|r_i - r_j| + |c_i - c_j|) \leq m \right\}, \quad (4)$$

with $m = \text{round}(2/0.45) = 4$ grid steps, computed as m steps of 4-connected dilation. Its negatives are all fire-adjacent. Four evaluations use

365 it (Table S1). A scores out-of-fold predictions from 5-fold blocked cross-validation ($k = 10$) on V_R , and B scores the same predictions on H_K . B is thus the **same blocked model restricted** to the scar area, which isolates the evaluation region from the training regime. C is **leave-one-scar-out**: a model fitted on $V_R \setminus H_K$ is scored on H_K , skipped if either set has one
 370 class. D, the **foreign-region evaluation**, averages over the four other regions a model fitted on that region’s population and scored on H_K . Buffers of 2, 5 and 10 km were run, 2 km primary. Three controls reuse A’s predictions, prevalence-matched, negative-pool and positive-pool, and a within-region half-split completes the set; their specification is in Section S1.9.3.

375 **Distance collars.** With 0.45 km per grid step on both axes, the distance to the nearest burned cell and the collar of radius r , for r of 5 and 10 km, are

$$d_i = 0.45 \min_{j \in P_R} \|(r_i, c_i) - (r_j, c_j)\|_2, \quad F_R(r) = \{i \in V_R : d_i \leq r\}. \quad (5)$$

Every burned cell has $d_i = 0$ and is kept, so only far-field negatives leave. In transfer the model is fitted on $F_s(r_s)$ and scored on $F_t(r_t)$, with the pairs
 380 of Table S8 and $r = \infty$ for the region-wide frame.

Every collar frame, H_K and $F_R(r)$ alike, is defined from burned cells, so it is label-conditioned. It is a diagnostic of how the scoring extent shapes the metric. It cannot be drawn before a fire.

3.13. Leakage control and reproducibility

385 An explicit forbidden-column set is enforced at every model fit as an assertion rather than a convention: coordinates and their normalised forms, every burn-date and label-provenance column, and the agreement fraction

are excluded from all feature sets. The natural-vegetation mask defines the population and is never a predictor. All randomness uses seed 42 and the bootstrap 1000 replicates (the diagnostic bootstraps of the Supplementary Material use per-measure offsets from that seed, so that independent measures do not share a draw). Across five seeds every transfer verdict at 1 km blocking is stable; at 5 km one level verdict and two paired-delta verdicts are not, and Section S1.21 identifies them. The repository’s own reproduction check refitted every within-region model and all twenty directed CORAL transfers against the frozen upstream output, Manavgat’s re-frozen one included (Section 3.2): the within-region comparisons agree exactly and the transfer directions to within 1.3×10^{-8} . The tolerances, the re-freeze’s one-line patch and the frame-transfer script’s run-to-run variation are in Section S3.6.2. The re-freeze and this check were carried out by the manuscript authors rather than independently by the pipeline’s original author. Headline results are repeated across two populations, three block sizes, the CORAL sweep, both feature sets and four classifier capacities, and where a conclusion depends on one of those choices **the dependence is reported rather than resolved by choosing the favourable setting** (Sections S1, S3.6).

3.14. Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla admits a comparison in which place is held fixed and the event varies: a second fire burned inside the identical AOI, on the identical grid, eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell bootstrap used elsewhere. **Season, year and population all differ**, since the 2022 arm is defined by removing the 2021 scar (Sections S1.13, S3.3).

4. Results

4.1. Study regions and the admissibility gate

415 All five candidate regions pass the burned-landcover gate, and the negative control fails it as intended. Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded. The five admitted regions carry 0.723 to 0.991, so the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the gate separating
420 natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, **though one negative control cannot establish it**. North Evia is analysed on an extended AOI. That leaves the burned scar essentially unchanged while cutting TSG prevalence from 0.676 to 0.287; the effect on transfer is in Section S1.1.

4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked out-of-fold ROC-AUC in every region. The increment's bootstrap interval excludes zero in all five regions at 1 km and at 5 km blocking (Fig. 3), the two scales this design supports as intervals. At 10 km the point estimates
430 hold, from +0.047 to +0.154. They rest on 6 to 33 positive-carrying blocks, however, and are indicative.

Table note (resampling units). The 20-cell row is indicative (above). The 10-cell row, with 16 to 70 positive-carrying blocks in every region, is the coarsest blocking this design supports properly. The block counts and the
435 coverage argument are in Section S1.3, *Table 1 note (resampling units)*.

The Manavgat rows are computed on the corrected label (Section 3.2). The rows strengthen rather than weaken; the change row by row is in Section

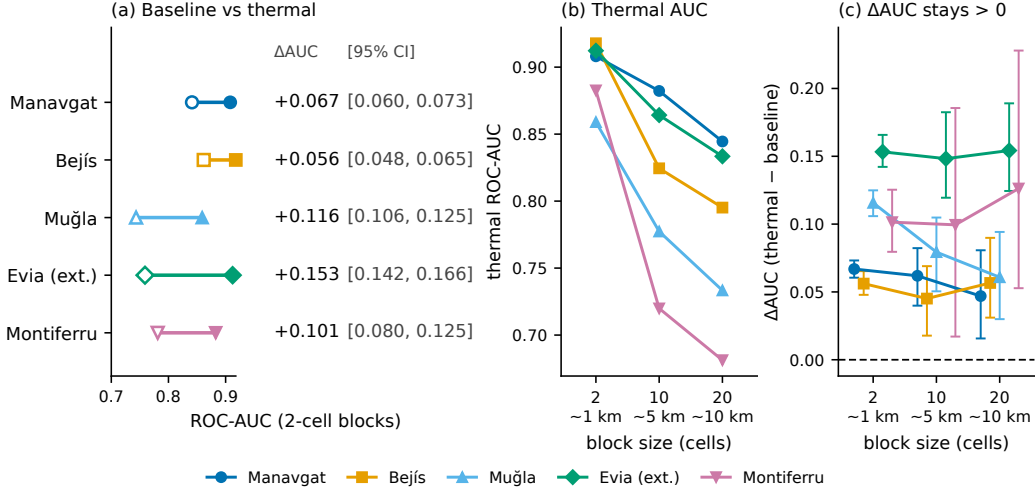


Figure 3: The thermal increment replicates in all five regions and survives coarser spatial blocking. Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, $\approx 1, 5$ and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region–block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 1, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge.** Colour and marker shape both encode region, so the panels remain readable in greyscale.

Table 1: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km. Baseline and Thermal columns are ROC-AUC; the 95 % CI belongs to Δ AUC.

Region	Block	Baseline	Thermal	Δ AUC	95% CI
Manavgat 2021	2	0.841	0.908	+0.067	[+0.060, +0.073]
	10	0.820	0.882	+0.062	[+0.040, +0.082]
	20	0.798	0.845	+0.047	[+0.016, +0.081]
Bejís 2022	2	0.862	0.918	+0.056	[+0.048, +0.065]
	10	0.779	0.824	+0.045	[+0.018, +0.069]
	20	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2	0.743	0.859	+0.116	[+0.106, +0.125]
	10	0.698	0.777	+0.079	[+0.050, +0.105]
	20	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2	0.759	0.912	+0.153	[+0.142, +0.166]
	10	0.716	0.864	+0.148	[+0.119, +0.182]
	20	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferru 2021	2	0.781	0.883	+0.101	[+0.080, +0.125]
	10	0.620	0.720	+0.099	[+0.017, +0.186]
	20	0.555	0.681	+0.126	[+0.053, +0.228]

S3.1.3.

4.3. What the evaluation frame is worth, within one region

440 This section isolates, on one model in one region, the effect Section 4.4 applies between regions. No transfer is involved, so nothing the transfer arms do can explain it (Section S1.9).

Where the skill is lost, on a matched comparison. Four evaluations are reported (Table S1). The last three are scored on **identical** 445 **cells**, so they differ only in what the model was trained on. The first shows why an unmatched comparison misleads. The held-out unit is a burned connected component of at least 50 cells, with all cells within 2 km of it. What makes that harder than a whole region is the composition of its negatives, not its burned fraction. Matching prevalence changes nothing, at 450 -0.002 $[-0.005, +0.001]$. Scoring the same predictions on the scar area

costs **+0.149** [**+0.087**, **+0.211**], and replacing only the negatives costs **0.139** [**0.098**, **0.181**], the whole of it. **The effect is the negative pool.** The single-pool controls are in Section S1.9.1.

All four rows are means over the **same seven scars**, four in Muğla, two
 455 in Montiferru and one in Evia, with Student t intervals over the scars. **Row A is a region-level quantity repeated identically across a region's scars, so its interval is pseudo-replicated and should not be read as coverage.** Clustering by region, over three regions, gives **+0.137** [**+0.048**, **+0.226**] for $A - B$ and **+0.266** [**-0.022**, **+0.553**] for $A - C$. **On the**
 460 **region unit only the frame cost is established** (Section S1.9.2).

The same model, scored two ways on the same region, differs by **0.133 AUC**. Rows A and B are one model and one set of out-of-fold predictions. The only change is which cells they are scored on. That change alone costs **0.133** [**+0.059**, **+0.207**] of the 0.227 [**+0.147**, **+0.308**] fall from
 465 A to C, about three fifths. That is a ratio of two estimates, so the point is the size of the numerator rather than the precision of the fraction. It is the size of the predictor-block increments this design itself measures. A region-wide blocked figure is therefore an upper bound on what the same model achieves where the fire actually is.

Neither withholding the fire nor moving 2,800 km has a measur-
 470 **able cost**. B minus C, the fire-specific residual on identical cells, is **+0.094** [**-0.012**, **+0.200**]. C minus D, the effect of replacing a same-region model with one fitted 306 to 2,802 km away, is **-0.007** [**-0.070**, **+0.057**]. Both span zero and both arms sit close to chance. **With seven scars, two of**
 475 **them starved of positives, this design cannot establish a fire-specific residual, only bound it at about 0.20.** Nor does it establish the fire *event* as the unit, since the held-out patch is defined by the labels and its identity

cannot be separated from its location. Section S1.23 reports the spread behind row D, and the sweeps showing that the patch definition does not drive
480 the result.

The increment declines with the holdout, and is not established once the fire is withheld. Contribution 2 concerns the paired thermal-minus-baseline difference, so the same evaluations were run on it. It is +0.056 to +0.153 under blocked cross-validation, with every interval above zero. It
485 is +0.028 on a within-region half-split, **+0.024** [**-0.040**, **+0.089**] under leave-one-scar-out, and +0.007 [-0.021, +0.037] across regions. The point estimate falls monotonically as the holdout hardens, and the last two intervals span zero. **The within-region increment is therefore substantially a property of interleaved holdout.**

490 4.4. *The same effect between regions, applied to our own matrix*

Section 4.3's effect applies with equal force between regions. This test is reported before the transfer matrix because it changes what Section 4.5 and Section S1.16 can claim. It withdraws nothing in Section 4.3, which is what it is built from. Sources, code and the elaboration are in Section S1.20.

The five areas of interest are not comparable frames. Each is a
495 rectangle drawn around a fire, and they differ by an order of magnitude in how much unburnt far field they enclose. The share of modelled cells beyond 10 km of any burned cell runs from **2.1** % in Montiferru to **63.1** % in Bejís (Table S13). That far field is not neutral. In Manavgat the median elevation
500 of modelled cells rises from 330 m within 5 km of the fire to 1,273 m at 20 to 50 km, against 287 m for the burned cells themselves.

Under an equalised frame four regions agree, and Manavgat does not. Restricting every region to cells within 10 km of any burned cell removes only far-field negatives. On that frame the other four regions agree

505 in sign on elevation, LST and TVDI, and Manavgat sits on the other side of 0.5 on all three. Its elevation reversal survives with support: 0.376 [0.300, 0.465] against Muğla's 0.606 [0.525, 0.685] and Evia's 0.648 [0.550, 0.740]. Under this paper's own criterion from Section 3.10, **two between-region reversals therefore remain bootstrap-supported under the collar**,
510 both on elevation and both involving Manavgat. Its thermal channels move toward 0.5 without reversing with support (current LST 0.522 [0.451, 0.591]). **The honest statement is that the collar removes every elevation reversal except Manavgat's, which it leaves supported.** The difference instrument, the two scope limits and the supported-feature counts are in Section S1.20, *Under an equalised frame four regions agree, and Manavgat does not.*

The sign most regions share is not the one the dryness framing predicts. At the point estimates a hotter pre-fire surface burned *less* in four of five regions. On mutual adjustment temperature survives where green-
520 ness does not, so the absolute channels behave there as static land-surface descriptors rather than as dryness. Manavgat is the exception. Its burned cells are hotter (current LST 0.665 on the full frame), and they are also low-lying. Within distance to the nearest burned cell its LST signal falls to 0.454, so the thermal sign there cannot be separated from its terrain gradient. In Montiferru the signal is attenuated to near-null. The stratifications,
525 the correlations and the reciprocal adjustment are in Section S1.11.

The same test weakens a result of our own, but no longer dissolves it. Under the frozen label, the diagnostic that best ordered transfer in our matrix reached $\rho = +0.84$. That diagnostic is the sign-agreement fraction over interval-supported features. It is built from exactly these signed
530 AUCs and was correlated against transfer on the same unequal frames. Un-

der the corrected label it no longer orders transfer on the frames as drawn
 $(\rho = +0.52 [-0.27, +0.87]$; Section S4). Recomputed under the collar it
is no longer degenerate. It takes the values 0 and 1 across sixteen defined
535 directions and correlates with collar transfer at $\rho = +0.38$ ($p = 0.15$). The
correction that once made it unanimous now leaves it weak on both frames
(its cosine variant and sources in Section S1.20).

Two further arms move with the frame. The same-geography arm
is the most extreme case in the cohort: its 2022 arm has 93.2 % of cells
540 beyond 10 km of any burned cell, against 55.3 % for 2021. **Only elevation
and slope carry a verdict there, on both the Muğla reversal is
removed, and on the thermal channels the arm is silent.** It rests on
eleven positive-carrying blocks against a floor of sixteen (Section S1.20). The
within-region increment, by contrast, **survives**, positive in all five regions at
545 a mean of +0.083 against +0.087 as drawn.

The transfer matrix moves as well (Table S8), from 0.527 as drawn
to 0.589 on the 10 km collar. A data-provenance defect in this arm was found
and corrected; every arm here reads each region’s predictor file by its recorded
SHA-256 (Section S1.20). **The baseline control must be restated on**
550 **this frame.** The static baseline transfers at 0.565 against 0.589, a paired
difference of +0.024 rather than +0.007. The control therefore holds in kind,
but the gap is about three times larger once frames are comparable, and
about twice as large at the 5 km collar.

What this settles, and what it leaves standing. Four quantities
555 reported below are properties of the frames rather than of the predictor-
burning relationship, and are identified as such where they appear. They
are the count of below-chance directions, which falls from seven to five; the
sign-agreement diagnostic; the same-geography arm; and the paired ther-

mal contribution, +0.007 as drawn against +0.024 equalised. One is not:
 560 Manavgat’s elevation reversal, which the collar leaves supported. The paired
 contribution carries the portability null, so it is given an interval on the frame
 this section argues for: **+0.024** [**−0.004**, **+0.049**] under the pair-cluster
 resampling of Section S1.15. It still spans zero, so the null survives the cor-
 rection, but only just. Under the alternative admissible unit, clustering by
 565 target region, it does not ([+0.011, +0.040]; Section S1.20). What survives
 is the central negative result, and its size must be stated on a matched com-
 parison. Setting 0.589 against a within-region reference of about 0.90 would
 compare a collar number with a full-rectangle one at 1 km blocking. Re-
 computed on the same frame at 5 km blocking, that reference is 0.786. The
 570 shortfall is therefore **+0.197** [**+0.091**, **+0.303**], a Student t interval over
 the five target regions rather than the spatial-block bootstrap used elsewhere.
 Manavgat contributes the largest per-region shortfall to it (+0.319), and the
 unmatched comparison of 0.29 is in Section S1.20. Section S3.5(ix) records
 the frame as a limitation of this cohort.

575 4.5. Cross-region transfer, and what label-free adaptation does to it

Everything in this section is computed on the frames as drawn and should be read against Section 4.4. The as-drawn matrix is re-
 ported because it is what the original analysis protocol yields. Per-direction
 values are in Table S16, and the supporting arms in Section S1.21.

580 **Raw transfer is heterogeneous and includes anti-predictive di-
 rections.** Target AUC spans 0.314 to 0.677, and Fig. 4 gives the matrix
 direction by direction. The support counts depend on the blocking scale, so
 both are reported. At 2-cell (≈ 1 km) blocking, eleven of twenty directions
 are above chance with interval support and seven below. At the 10-cell (≈ 5
 585 km) blocking this design defends, nine are above and six below with five un-

certain; these are the counts Table S8 uses. Of the below-chance directions, two keep interval support after frame equalisation, Manavgat to Bejís and Muğla to Manavgat. Seed stability is in Section S1.21. Even the best raw transfer sits far below the target’s own within-region skill: Evia to Manavgat reaches 0.677 against Manavgat’s 0.908. The raw deficit runs from 0.231 to 0.594, against a reference that Section 4.3 shows is not matched to a transfer evaluation.

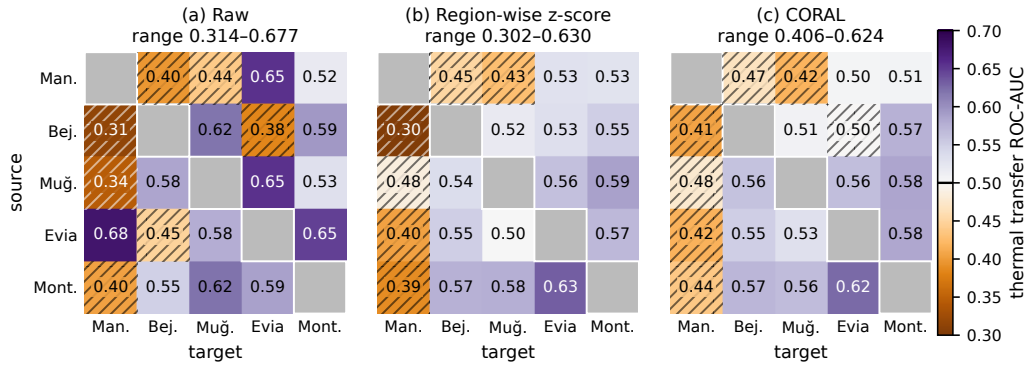


Figure 4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. (a) Raw transfer, no adaptation. (b) After region-wise z-score standardisation. (c) After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation narrows the spread from 0.314–0.677 (raw) to 0.302–0.630 (z-score) and 0.406–0.624 (CORAL). It compresses the matrix toward chance rather than lifting transfer, with one exception at the low end: under z-scoring Bejís→Manavgat moves further below chance (0.314 to 0.302). Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so three cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance), while z-score Evia→Muğla is 0.5013 and CORAL Manavgat→Evia 0.5043 (both un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.6:1). Confidence intervals for these directions are given in Table S16. Manavgat directions are on the corrected label (Section 3.2).

Manavgat is the weakest target in the matrix, at a mean of 0.435 over its four incoming directions against 0.494 to 0.572 for the other targets. It is not the weakest source (0.502, against 0.477 for Bejís). Three of its incoming directions lie between 0.314 and 0.404, while Evia to Manavgat reaches 0.677. An exploratory split by fire phase suggests where the collapse sits; it was not registered, and it concerns one region and one event. The corrected label’s added cells burned on 28–31 July, the fire’s first four days, and the remaining 784 later. Transfer into the early cells is lower than into the later ones in every direction: Bejís 0.276 against 0.419, Montiferru 0.351 against 0.549, Muğla 0.332 against 0.382, and Evia 0.667 against 0.704. The early cells also lie far lower, at a median of 219 m against 512 m for the later burns and 1,004 m for unburned cells (`paper/labelfix_rerun/round5/s4b_*`). Section 5.4 treats this as a mechanism proposal, not as evidence.

In precision terms it is worse than the ROC figures suggest. A susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity. PR-AUC averages **0.181 against a no-skill baseline of 0.157. Seven of twenty directions fall below their own baseline at the point estimate, all seven with intervals entirely below it.** Only one direction, Evia to Manavgat, exceeds twice its baseline. These are frame-as-drawn quantities, and the PR arm was not recomputed on the collar (Table S11).

The static baseline does not transfer either, at a mean of **0.519** against **0.527** for the thermal model. The failure is therefore not specific to the dynamic block. A baseline that does not travel, plus pre-fire thermal state, gives a model that does not travel. Section 4.4 restates this control on the equalised frame, where it is 0.565 against 0.589.

The thermal block’s paired contribution is +0.007, and the

620 **mean is not the informative statistic.** The individual contributions span -0.148 to $+0.133$, twelve positive and eight negative. The largest is about twenty times the mean, and the sign belongs to the pair rather than to the block. A mean near zero records cancellation, not consistent absence of effect. The directions are not independent, since each region appears in
 625 eight of the twenty, so the interval depends on the resampling unit. **All four units the design permits give the same answer**, from $[-0.018, +0.033]$ to $[-0.028, +0.045]$, and none propagates within-direction sampling variability. A leave-one-region-out jackknife never reverses its sign (Section S1.21).

630 **Label-blind adaptation compresses the matrix toward chance rather than repairing it** (Fig. 5). Under region-wise z-scoring the twenty directions span 0.302 to 0.630, and under CORAL 0.406 to 0.624. Sixteen of twenty move closer to chance, which helps the directions that failed and harms those that worked. Closer is measured as distance from 0.5, so it does
 635 not mean the matrix stays on one side of chance. Evia to Manavgat overshoots, falling from 0.677 to 0.404 under z-scoring and 0.417 under CORAL, below chance on the other side. The committed-in-advance CORAL arm averages **0.517**. Taking whichever method scores better per direction gives 0.523, but that selection uses the target labels the protocol forbids. It is
 640 therefore **an oracle upper bound rather than an achievable result**. Even the oracle stays below the reference a model reaches on an unseen scar (Table S1, row C, 0.546). A sign reversal is not a distribution mismatch that realigning inputs would repair. Section S1.10 reports the recovery fractions, including the seven directions with *negative* recovery, five of them with
 645 intervals entirely below zero.

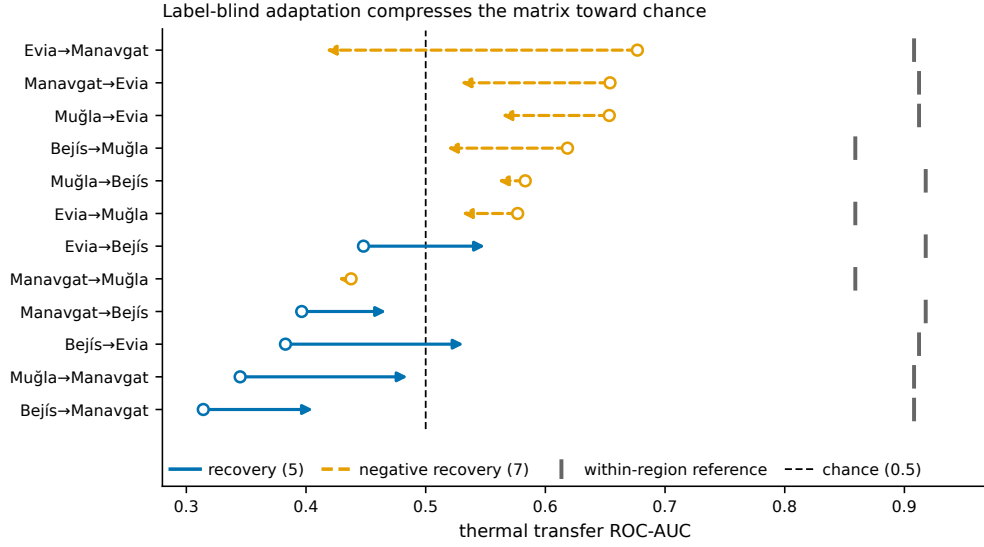


Figure 5: **Label-blind adaptation compresses the matrix toward chance.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves away from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. All six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so eleven of twelve end closer to chance and adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.438 and is pushed further down to 0.427. Bejís→Manavgat is drawn as a recovery because its better method, CORAL, lifts it from 0.314 to 0.406; its z-score arm alone moves further below chance, to 0.302 (Fig. 4). The worst case is Evia→Manavgat, where raw transfer of 0.677 is dragged down to 0.417 against a within-region reference of 0.908, a recovered fraction of -1.126 . No arrow reaches its within-region tick. Manavgat directions are on the corrected label (Section 3.2).

4.6. Further arms

Six further arms bear on the findings above without changing them. They are the contrast pair (Section S1.16, Fig. 6), the two interventions (Section S1.14), the sensitivity summary (Section S1.17), the same-geography
650 two-event arm (Section S1.13), the distance curve (Section S1.18) and the label-budget curve (Section S1.19). Each is stated there with its own limits. Pooling every other region does not beat the best single source for four of five targets; for Evia the pooled model exceeds it (0.715 [0.668, 0.757] against 0.654; Fig. 8). Removing the direction-reversing features costs -0.076 within
655 region and returns nothing measurable on transfer, $+0.014$ [-0.028 , $+0.056$] (Fig. 7). The two removed features were fixed from the frozen label's supported reversals, and are kept, not re-selected. Thirty-two labelled 5 km blocks recover 83 to 89 % of the target's matched ceiling in three of six directions, and 30 to 52 % in the rest. Those blocks are 7 to 20 % of the target's
660 population, drawn from the event being predicted, which is not a resource available before that event burns.

No diagnostic among the twenty fixed in advance orders transfer with an interval excluding zero (Section S4). One of them, the supported-feature vector Spearman, is defined on only six directions and was not evaluated as
665 a diagnostic.

5. Discussion

5.1. Reading the three findings together

Section 1.3 states the three findings and Section 4 establishes them; this section argues from them. The first is not a caveat attached to the second; it

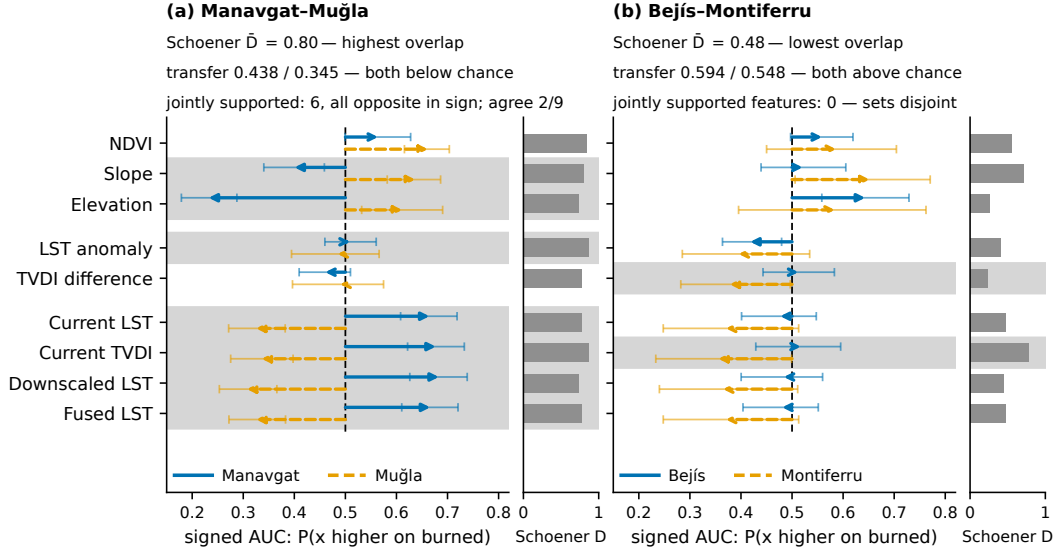


Figure 6: **The contrast pairs.** (a) Manavgat–Muğla, the pair whose burned cells overlap most in environmental space (mean Schoener $\bar{D} = 0.80$, rank 1 of 10 on all five overlap measures), transfers below chance in both directions at the point estimates (0.438 and 0.345). (b) Bejís–Montiferru, the pair that overlaps least ($\bar{D} = 0.48$, rank 10), transfers above chance in both directions at the point estimates (0.594 and 0.548). Arrows run from 0.5 to each region’s signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one. A filled head marks a 95% spatial-block (10-cell, ≈ 5 km) bootstrap interval that excludes 0.5; an open head marks one that does not. Solid blue is the first region named, dashed orange the second; line style is included because the two hues nearly merge in greyscale. Shaded rows mark features whose sign differs between the two regions, and side bars give Schoener’s D per feature. In (a), six features are interval-supported in both regions, and all six have opposite signs; the signs agree on 2 of 9 features overall. In (b) the two supported sets do not intersect, and the signs agree on 7 of 9. The transfer verdicts use the 1 km bootstrap, under which all four directions are off chance. At 5 km blocking only Muğla $\rightarrow$ Manavgat stays interval-supported, which is why the contrast is stated at the point estimates. Manavgat values are on the corrected label (Section 3.2).

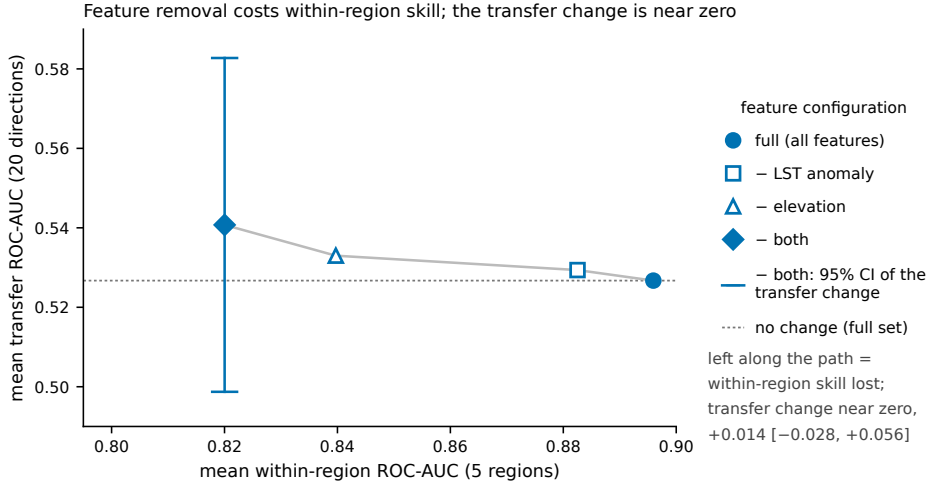


Figure 7: Removing the direction-reversing features costs locally and returns nothing measurable on transfer. Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, within-region skill falls monotonically ($0.896 \rightarrow 0.820$) while the transfer mean rises monotonically ($0.527 \rightarrow 0.541$). Both monotonicities are verified when the figure is built. The within-region cost is 0.076 AUC and is supported in every region. The change in transfer is near zero: +0.014 AUC, with a pair- t interval over the ten region pairs of $[-0.028, +0.056]$ that spans zero. No exchange between the two is demonstrated, since a local cost is measured and no compensating transfer gain is. The vertical whisker at the “without both” point is that interval, drawn about the full-set transfer level (dotted line, no change), so its crossing of the no-change level can be read directly. Manavgat values are on the corrected label (Section 3.2).

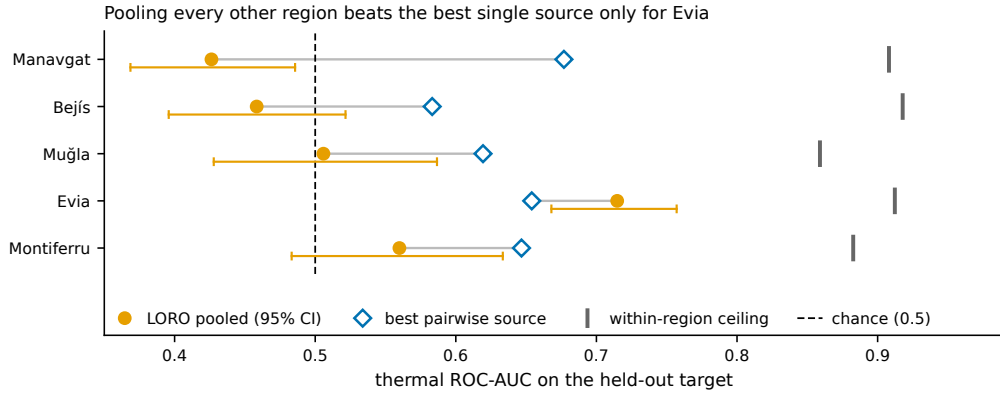


Figure 8: **Pooling every other region does not beat the best single source for four of five targets.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle, with its 95% spatial-block (≈ 5 km) bootstrap interval drawn just below the row) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in four targets. For Evia it lies to the right, at 0.715 [0.668, 0.757] against 0.654 for the best single source, Manavgat (Section 5.3). Elsewhere, adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.426, Bejís 0.458), and only for Manavgat does the interval lie entirely below chance, while a single well-chosen source reaches 0.677 and 0.583 respectively. Every point remains far below the within-region ceiling. Manavgat values are on the corrected label (Section 3.2).

670 is the instrument that sets its size. Applied to our own matrix, it identified four quantities as properties of the frames rather than of the relationship between predictors and burning. What it left standing is a shortfall in transferred skill and one reversed relationship, Manavgat’s elevation. The third closes the obvious way around the second. If transfer cannot be assumed, it 675 might still be anticipated from how similar two regions are, and no diagnostic in the set fixed in advance was shown to do that.

5.2. *Why the thermal increment is real but local*

The within-region increment and the transfer failure are measured at different separations, and Section 4.3 shows most of the difference is already 680 present inside a single region. It holds where held-out cells are interleaved with training cells, and most of it is gone once they are not, before the fire or the region changes. It is not an artefact to be explained away: it survives every sensitivity arm of Sections S1.1–S1.8, and window closure stays positive and supported in all five regions on the corrected label. But it is established 685 under interleaved validation and not beyond it. Even there the window-closure arm **weakens monotonically in Evia**, so what holds everywhere is survival, not improvement.

That comparison is established at the scar, not at the region, and this weakens the claim. On the region as the unit only the frame 690 cost is established; the fall to an unseen scar has scar-level support only (Section 4.3).

The natural objection is that Mediterranean regions are simply different systems, so a predictor meaning one thing in one place and another elsewhere compares two systems rather than showing instability. **We designed the 695 two-Muğla-events arm to answer that objection and it does not answer it:** Section 4.4 shows it is the most extreme frame artefact in the

cohort. Three further confounds were never resolved in any case. Season and year **cannot be resolved in this study area**, the population is not fixed, and the positive-block count is below this design’s own floor (Section 700 S3.5(ii)). With one fire per region everywhere else and that arm withdrawn, **this cohort provides no evidence that the transfer shortfall is regional rather than event-specific**. Section S1.18 adds a length scale for the within-region decay but cannot turn it into an attribution either.

5.3. *What the two interventions do and do not show*

705 Both interventions show a similar shape, with one exception. Pooling four regions does not beat the best single-source transfer for four of five targets, and it stays well below the within-region ceiling for all five (Section S1.14). For Evia the pooled model exceeds the best single source, at 0.715 [0.668, 0.757] against 0.654. An observation on that pairing is in Section S1.14.1. 710 Elsewhere pooling does not manufacture the missing conditional information. Removing the reversing predictors costs -0.076 of within-region skill with interval support in every region, and returns $+0.014$ $[-0.028, +0.056]$ in mean transfer.

That pair of numbers is easy to read as an exchange, and it is not one. 715 Both arms are null on the portability axis. The thermal block’s own contribution is $+0.007$ and removal returns $+0.014$, both with intervals spanning zero, so they measure a local cost and no compensating gain. That is not a conservation law and not a rate at which local skill can be sold for portability; no such rate is estimated here.

720 5.4. *The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data confirmed the null. The regime-distance correlation has the wrong sign at the

point estimate ($\rho = +0.109$). The most regime-similar pair, Bejís and Manavgat, fails in both directions (0.396 and 0.314). The most regime-different pair, Bejís and Muğla, transfers above chance in both (0.618 and 0.583). One mundane explanation can be set aside: subsampling Muğla, much the largest population, to Manavgat’s cell count leaves its transfer behaviour inside the subsampling range in both roles. The error was in the hypothesised grouping, not in the data, and with ten pairs this cannot refute regime typology (Archibald et al., 2013) in general.

Manavgat is where the matrix fails most, and three indicators point there. It has the most outlying univariate profile in the cohort, with elevation at 0.232 and the three LST channels near 0.67, all interval-supported and all opposite to the other four regions. It also carries the largest matched shortfall (+0.319) and is the weakest target (0.435). A model trained where burned cells sit higher and cooler than their surroundings is asked to rank a fire that burned low and hot. In the exploratory phase split of Section 4.5, the cells that burned in the fire’s first four days lie at a median of 219 m, and their LST signal is the strongest in the region (AUC 0.70 to 0.72 with interval support, against 0.57 to 0.58 later). Transfer into these early cells is lower from every source, and the static baseline collapses there too. Elevation is reversed in both phases (0.197 and 0.328). With this data, elevation and temperature cannot be separated, because the low ground is also the hot ground.

We therefore offer a mechanism proposal, not a finding. The sign of elevation, and with it the sign of the thermal channels, may be set by which part of the elevation gradient an event burns relative to its frame, rather than by the region. Muğla shows the same pattern across its 2021 and 2022 events. The 2021 fire burned high relative to its frame (elevation AUC 0.611), the

750 2022 fire low (0.297), and the reversal is interval-supported inside a single study area. Muğla’s reversal is carried by the 2022 arm’s far field and the collar removes it (0.606 against 0.565), whereas Manavgat’s survives the collar (0.376). The two cases therefore share the pattern on the frames as drawn but differ in how much of it the frame explains. Testing the proposal needs
755 several events per region spread along the elevation gradient, and predictors that separate terrain from surface temperature.

Measures built from interval support carry a warning for future work. A measure that counts only interval-supported features changes its supported set when one bound sits on the threshold, and one such flag moved
760 the full-frame supported-feature cosine from $\rho = 0.49$ to 0.70. The flag was Manavgat’s NDVI, at $[0.499, 0.628]$ in the pipeline’s bootstrap and $[0.502, 0.624]$ in ours, and under both labels five to six of the pipeline’s bounds lie within 0.01 of 0.5. This partly explains why the conditional-similarity result fell from +0.84 to +0.52 $[-0.27, +0.87]$ under the corrected label. Work that
765 uses such measures should report how many bounds lie near the threshold, and should repeat the calculation across bootstrap streams before reading a correlation.

5.5. *The empirical contrast with Dimarco et al.*

Dimarco et al. (2026) transfer successfully across a comparable Mediter-
770 ranean design and we do not, and the two results are not in conflict. Their predictors are attributes of a place, ours the state of a surface in one season. That suggests the relation between domain similarity and transfer success depends on the predictor class.

That reading is one of at least two, and we cannot separate them here.
775 Their response variable is human-driven **ignition**, dominated by access and activity, while ours is burned **area**, dominated by spread. A predictor-class

explanation and a response-variable explanation are therefore confounded in this comparison. Our own data speak against a simple predictor-class reading in any case: the static baseline transfers at a mean of 0.519 here, so within
780 this cohort the place-attribute class does not travel either.

5.6. Implications

In precision-recall terms, which is how a susceptibility surface is used, a model moved to a region it was not fitted in does not usefully rank burned cells there (Section 4.5, **on the frames as drawn**).

785 The paper supports one concrete change in reporting: alongside a spatially blocked within-region figure, report skill on a held-out burn scar and its surroundings. On these five regions the two differ by about 0.13 ROC-AUC on the same model, the size of the effect such papers usually claim, and this frame cost holds with the region as the unit (Section 5.2). **A blocked**
790 **figure alone should therefore be read as an upper bound.** Transfer skill likewise has to be *measured* on the target region rather than assumed from a within-region figure. Where a model must be moved, the resource that closes the gap is target labels (Section S1.19): a real answer, but not a cheap one.

795 The results also bound what unsupervised alignment can be asked to do. Even the oracle selection only reaches the reference a model can reach on an unseen scar (Section 4.5). Alignment is not failing far below an achievable target; it is regressing the matrix onto it, at the cost of the directions that already worked. The compression is not uniform: under z-scoring Bejís to
800 Manavgat moves further below chance, from 0.314 to 0.302, and Evia to Manavgat crosses to the other side of it (Section 4.5). A sign reversal is not a distribution mismatch that realigning inputs repairs.

5.7. Limitations

Section S3.5 states the limitations in full. Seven bind the conclusions
805 above.

- **The frames are not comparable and this cohort cannot fully repair it**, since frames fixed upstream can be restricted but not extended; fixing them by an accessible-area rule (Barve et al., 2011) before any predictor is computed is the main design lesson of this paper
810 (Section 4.4; S3.5(ix)).
- **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology and the shortfall cannot be attributed to region rather than event (Section 5.2; S3.5(v)).
- **The diagnostic correlations rest on an effective sample of ten region pairs**, because the two directions of a pair share both regions,
815 so the results of Section S4, successes and failures alike, read at that power (S3.5(xiii)).
- **The interval-support measures are less stable than the point estimates behind them**, since five to six bounds lie within 0.01 of 0.5
820 and one flag can move a diagnostic correlation by about 0.2, whereas the transfer verdicts are steadier across bootstrap seeds (Section 5.4; Section S1.21; S3.5(viii)).
- **Manavgat’s atypical transfer is localised, not explained**: its meteorology, the quality screening and the frame were tested and none
825 accounts for it, and the localisation rests on one region and one event (S3.5(vii)).

830 • **The phase split cannot separate elevation from temperature**, because the cells that burned first are both the lowest and the hottest, so the proposal of Section 5.4 is confounded by construction and was not registered (S3.5(xi)).

- **The quality-screening arms differ in code version as well as in screening**, yet agree on elevation to four decimals and on every other signed AUC to within 0.005, so the confound could hide an effect only if two cancelled (Section S1.5; S3.5(xii)).

835 The remaining six, from the absent meteorological covariates to the classifier comparison made by point estimate only (Section S1.8), bear on scope rather than on the conclusions above.

- **No meteorological covariates** enter the models, so we cannot say how local skill and portability behave for a mixed thermal-plus-weather predictor set (S3.5(i)).

- **The same-geography comparison covers one region only**, where year and seasonal phase are confounded and the 2022 arm has eleven positive-carrying blocks against a floor of sixteen (Section S1.13; S3.5(ii)).

- 845 • **All labels derive from a single burned-area product**, MCD64A1 (Giglio et al., 2018), whose omission and commission characteristics (Boschetti et al., 2019) bound every model here, and no second product covers both years at this resolution (S3.5(iii)).

- 850 • **Evia remains the most imbalance-atypical population** even on the extended AOI, with a TSG prevalence of 0.287 against 0.070 to 0.212 in the other four regions (S3.5(iv)).

- **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions, so exact reproduction requires the archived environment to which every reported number is fixed (S3.5(vi)).
- **Other classifiers are compared by point estimate only:** three further estimators land between 0.481 and 0.527 with eleven to thirteen of twenty directions above chance, and their ordering is not claimed (Section S1.8; S3.5(x)).

6. Conclusions

Where a fire model is scored decides what it appears to know. Model, predictors and fitting were held fixed, and only the scored cells changed. Moving from a whole study region to the burn scar and its 2 km collar costs **0.133 ROC-AUC**. A control measures the cause as the composition of the negative pool rather than class balance. That is the size of the predictor-block increments this design itself measures. A region-wide validation figure should therefore be read as an upper bound on what a model achieves where fire actually occurs, and reported as one.

That correction is not only other people's problem. Applied to our own five-region matrix, it shows four of our own quantities to be properties of the frames. The study areas enclose very unequal far fields. Equalising them to a 10 km collar lifts mean transfer from 0.527 to 0.589 and reduces the directions below chance. It removes every supported elevation reversal but one, Manavgat's, which survives. The same correction applies to the one arm that held place fixed, two fires in one study area eleven months apart. We had wrongly exempted it, because its geography was constant while its evaluation frame was not. Its elevation reversal does not survive equalisation,

and that arm can speak only for the year-invariant channels. Much of what had looked like a reversed predictor-burning relationship was a statement
880 about how five rectangles were drawn.

What survives is the central result. Pre-fire thermal predictors add a real and repeatable within-region increment in all five regions. It survives spatial blocking at about 5 km, and a predictor window closed up to two weeks before the first labelled burning. It does not travel. On matched frames and
885 blocking, equalised transfer of 0.589 falls 0.197 short of the within-region reference. The paired cross-region contribution spans zero on both frames under the resampling unit we treat as primary, though not when clustered by target region (Section 4.4), and its sign varies by pair. The static baseline transfers no better on either frame, so this is not a peculiarity of thermal
890 predictors. Label-free alignment compresses most directions towards chance rather than repairing them.

Two limits belong with all of this. The thermal direction shared by most regions is the opposite of the one the dryness framing predicts. Hotter pre-fire surfaces burned less in four of five regions, and holding greenness does
895 not remove it, while holding temperature reverses greenness in two regions. On this cohort the absolute channels therefore behave as static land-surface descriptors rather than as a dryness index. Manavgat is the exception, and there its thermal sign cannot be separated from its terrain. And with one fire per region, and the single arm that held place fixed now withdrawn, this
900 design measures a transfer shortfall but cannot attribute it to region rather than to event.

Three practical consequences follow. Report local skill and portability separately, since a predictor block worth a large within-region increment may contribute nothing distinguishable from zero across regions. Fix the

905 evaluation frame by an explicit accessible-area rule before any predictor is
computed, because a frame chosen for convenience can manufacture both a
transfer failure and a mechanism to explain it. And do not screen for transfer
by similarity. None of the twenty diagnostics fixed in advance was shown to
order transfer. At the point estimates, the most similar pair failed in both
910 directions while the least similar transferred above chance. Without labels
from the target region, a model's skill there is unknown, and no measure
computed beforehand stands in for it. With them, skill can be measured
and the model recalibrated, but the price depends on the direction. Thirty-
two labelled 5 km blocks recover 83 to 89 % of the target's matched ceiling
915 in three of six directions, and only 30 to 52 % in the other three (Section
S1.19). Those labels came from the event being predicted, so the curve
prices the gap rather than offering a way to close it before a fire. Whether
labels from earlier fires in the same region would serve is not tested here.
Three extensions are left for future work: temporal transfer, meteorological
920 covariates, and physically normalised dryness variables.

Declarations

Data and code availability

All satellite inputs are public and were retrieved through Google Earth
Engine: burned-area labels from MODIS MCD64A1 Collection 6.1, land
925 cover from ESA WorldCover, and the thermal, optical and terrain inputs
described in Section 3.4. No proprietary or restricted data were used.

The upstream processing pipeline, `satellite-thermal-digital-twin`
(E. Metin), is public at <<https://github.com/emrehann17/satellite-thermal-digital-twin>> under the MIT licence. The commit of record for every number
930 reported here is `6381f4c`. The modelling dataset behind each number is

identified by SHA-256 in the pipeline’s canonical-input record, and every analysis script verifies that hash on load.

The analysis code, the five modelling datasets, the frozen numeric outputs and the Supplementary Material are at <<https://github.com/dryuemco/thermal-twin>>. Fig. 1–8 and the graphical abstract are drawn by the scripts in `paper/figures/`, which also assert Table 1 and every plotted value against the frozen outputs. Sixteen of the twenty-two supplementary tables (S2–S7, S9–S18) are rebuilt row by row by `paper/code/appendix_tables.py` from source files whose SHA-256 values are pinned in `paper/labelfix_rerun/round5/tables/SOURCES.sha256`; the others name their source files in their captions. `paper/figures/check_all.py` runs all of these checks. Software: Python 3.12.10 with NumPy 2.4.4, pandas 3.0.2 and scikit-learn 1.9.0; the cross-region tolerance between scikit-learn versions is about ± 0.02 to 0.03 (Section S3.5(vi)).

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Competing interests

The authors declare no competing interests.

Ethics approval

960 Not applicable. This study involved no human participants, animal subjects, or personally identifiable data.

Author contributions

Stated in CRediT terms. **Emrehan Metin:** Software, Data curation, Investigation, Validation, Writing – review and editing. **Yunus Emre**
965 **Cogurcu:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic)
970 in order to write and run analysis and verification code against the frozen pipeline outputs, cross-check reported numbers against those outputs, and draft and edit manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

975 **References**

Alkan Akıncı, H., Akıncı, H., 2023. Machine learning based forest fire susceptibility assessment of Manavgat district (Antalya), Turkey. Earth Science Informatics 16, 397–414. doi:10.1007/s12145-023-00953-5.

- Archibald, S., Lehmann, C.E.R., Gómez-Dans, J.L., Bradstock, R.A., 2013. Defining pyromes and global syndromes of fire regimes. Proceedings of the National Academy of Sciences 110, 6442–6447. doi:10.1073/pnas.1211466110.
- Barve, N., Barve, V., Jiménez-Valverde, A., Lira-Noriega, A., Maher, S.P., Peterson, A.T., Soberón, J., Villalobos, F., 2011. The crucial role of the accessible area in ecological niche modeling and species distribution modeling. Ecological Modelling 222, 1810–1819. doi:10.1016/j.ecolmodel.2011.02.011.
- Boschetti, L., Roy, D.P., Giglio, L., Huang, H., Zubkova, M., Humber, M.L., 2019. Global validation of the Collection 6 MODIS burned area product. Remote Sensing of Environment 235, 111490. doi:10.1016/j.rse.2019.111490.
- Breiman, L., 2001. Random forests. Machine Learning 45, 5–32. doi:10.1023/A:1010933404324.
- Chuvieco, E., Cocero, D., Riaño, D., Martín, P., Martínez-Vega, J., de la Riva, J., Pérez, F., 2004. Combining NDVI and surface temperature for the estimation of live fuel moisture content in forest fire danger rating. Remote Sensing of Environment 92, 322–331. doi:10.1016/j.rse.2004.01.019.
- Dimarco, N.A., Faraji, I., Wahbi, M., Maatouk, M., Boulaassal, H., Yazidi Aalaoui, O., El Kharki, O., 2026. Generalizing human-driven wildfire ignition models across Mediterranean regions using harmonized remote-sensing and machine-learning data. Geomatics 6, 13. doi:10.3390/geomatics6010013.

- Gelabert, P.J., Jiménez-Ruano, A., Ochoa, C., Alcasena, F., Sjöström, J., Marrs, C., Ribeiro, L.M., Palaiologou, P., Bentu Martínez, C., Chuvieco, E., Vega-García, C., Rodrigues, M., 2025. Assessing human-caused wildfire ignition likelihood across Europe. *Natural Hazards and Earth System Sciences* 25, 4713–4729. doi:10.5194/nhess-25-4713-2025.
- Giglio, L., Boschetti, L., Roy, D.P., Humber, M.L., Justice, C.O., 2018. The Collection 6 MODIS burned area mapping algorithm and product. *Remote Sensing of Environment* 217, 72–85. doi:10.1016/j.rse.2018.08.005.
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202, 18–27. doi:10.1016/j.rse.2017.06.031.
- Huang, Y., Liu, N., Chen, W., Tan, W., Deng, Y., Wang, Z., Chlus, A., Shen, J., Townsend, P.A., 2026. Enhancing transferability of foliar trait retrieval models: A comparative analysis of transfer learning strategies and domain shift characterization. *Remote Sensing of Environment* 337, 115336. doi:10.1016/j.rse.2026.115336.
- Iban, M.C., Sekertekin, A., 2022. Machine learning based wildfire susceptibility mapping using remotely sensed fire data and GIS: A case study of Adana and Mersin provinces, Turkey. *Ecological Informatics* 69, 101647. doi:10.1016/j.ecoinf.2022.101647.
- Jodhani, K., Bhatiya, L., Sirvi, L., Chanda, A., Thakkar, S., Rathnayake, U., 2026. Artificial intelligence and remote sensing frameworks for wildfire monitoring and risk analysis across multiple ecosystems: A review. *AIMS Environmental Science* 13, 56–98. doi:10.3934/environsci.2026004.

- Kondylatos, S., Prapas, I., Camps-Valls, G., Papoutsis, I., 2023. Meso-geos: A multi-purpose dataset for data-driven wildfire modeling in the Mediterranean, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023) Datasets and Benchmarks Track, pp. 50661–50676. doi:10.52202/075280-2204.
- 1030
- Liu, C., Mostafavi, A., 2025. WildfireGenome: Interpretable machine learning reveals local drivers of wildfire risk and their cross-county variation. arXiv preprint arXiv:2511.11589. doi:10.48550/arXiv.2511.11589, arXiv:2511.11589.
- 1035
- Lobo, J.M., Jiménez-Valverde, A., Real, R., 2008. AUC: a misleading measure of the performance of predictive distribution models. Global Ecology and Biogeography 17, 145–151. doi:10.1111/j.1466-8238.2007.00358.
- 1040
- X.
- Maffei, C., Alfieri, S.M., Menenti, M., 2018. Relating spatiotemporal patterns of forest fires burned area and duration to diurnal land surface temperature anomalies. Remote Sensing 10, 1777. doi:10.3390/rs10111777.
- Maffei, C., Lindenbergh, R., Menenti, M., 2021. Combining multi-spectral and thermal remote sensing to predict forest fire characteristics. ISPRS Journal of Photogrammetry and Remote Sensing 181, 400–412. doi:10.1016/j.isprsjprs.2021.09.016.
- 1045
- Maffei, C., Menenti, M., 2019. Predicting forest fires burned area and rate of spread from pre-fire multispectral satellite measurements. ISPRS Journal of Photogrammetry and Remote Sensing 158, 263–278. doi:10.1016/j.isprsjprs.2019.10.013.
- 1050

- Meyer, H., Reudenbach, C., Hengl, T., Katurji, M., Nauss, T., 2018. Improving performance of spatio-temporal machine learning models using forward feature selection and target-oriented validation. *Environmental Modelling & Software* 101, 1–9. doi:10.1016/j.envsoft.2017.12.001.
- 1055
- Moreno-Torres, J.G., Raeder, T., Alaiz-Rodríguez, R., Chawla, N.V., Herrera, F., 2012. A unifying view on dataset shift in classification. *Pattern Recognition* 45, 521–530. doi:10.1016/j.patcog.2011.06.019.
- Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C., Thépaut, J.N., 2021. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* 13, 4349–4383. doi:10.5194/essd-13-4349-2021.
- 1060
- 1065 Pausas, J.G., Keeley, J.E., 2021. Wildfires and global change. *Frontiers in Ecology and the Environment* 19, 387–395. doi:10.1002/fee.2359.
- Persello, C., Bruzzone, L., 2012. Active learning for domain adaptation in the supervised classification of remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing* 50, 4468–4483. doi:10.1109/TGRS.2012.2192740.
- 1070
- Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin, A., Gourlet-Fleury, S., Péliissier, R., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications* 11, 4540. doi:10.1038/s41467-020-18321-y.
- 1075

- Podschwit, H., Jolly, W., Alvarado, E., Verma, S., Ponce, B., Markos, A., Aliaga-Nestares, V., Rodriguez-Zimmermann, D., 2022. Reliability of cross-regional applications of global fire danger models: A Peruvian case study. *Fire Ecology* 18, 25. doi:10.1186/s42408-022-00150-7.
- 1080 Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40, 913–929. doi:10.1111/ecog.02881.
- 1085 Rousseau, J.S., Betts, M.G., 2022. Factors influencing transferability in species distribution models. *Ecography* 2022, e06060. doi:10.1111/ecog.06060.
- Sandholt, I., Rasmussen, K., Andersen, J., 2002. A simple interpretation of the surface temperature/vegetation index space for assessment of surface moisture status. *Remote Sensing of Environment* 79, 213–224. doi:10.1016/S0034-4257(01)00274-7.
- 1090 Sun, B., Feng, J., Saenko, K., 2016. Return of frustratingly easy domain adaptation, in: *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence (AAAI-16)*, AAAI Press, Phoenix, Arizona, USA. pp. 2058–2065. doi:10.1609/aaai.v30i1.10306.
- 1095 Tuia, D., Persello, C., Bruzzone, L., 2016. Domain adaptation for the classification of remote sensing data: An overview of recent advances. *IEEE Geoscience and Remote Sensing Magazine* 4, 41–57. doi:10.1109/MGRS.2016.2548504.

- 1100 Valavi, R., Elith, J., Lahoz-Monfort, J.J., Guillerá-Arroita, G., 2019. blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution* 10, 225–232. doi:10.1111/2041-210X.13107.
- 1105 VanDerWal, J., Shoo, L.P., Graham, C., Williams, S.E., 2009. Selecting pseudo-absence data for presence-only distribution modeling: How far should you stray from what you know? *Ecological Modelling* 220, 589–594. doi:10.1016/j.ecolmodel.2008.11.010.
- Vesk, P.A., Morris, W.K., Neal, W.C., Mokany, K., Pollock, L.J., 2021. 1110 Transferability of trait-based species distribution models. *Ecography* 44, 134–147. doi:10.1111/ecog.05179.
- Vibhandik, P., Sawant, S., Joshi, A., Bidwe, R., 2026. A systematic literature review on forest fire susceptibility mapping using geo-spatial technology and future research directions. *Discover Artificial Intelligence* 6, 921. 1115 doi:10.1007/s44163-026-00921-0.
- Xu, Y., Dai, Y., Chang, L., Wang, Q., Dong, Y., 2026. Does your wild-fire prediction model actually work, or just score well? arXiv preprint arXiv:2605.18911. doi:10.48550/arXiv.2605.18911, arXiv:2605.18911.
- Yebra, M., Dennison, P.E., Chuvieco, E., Riaño, D., Zylstra, P., Hunt, E.R., 1120 Danson, F.M., Qi, Y., Jurdao, S., 2013. A global review of remote sensing of live fuel moisture content for fire danger assessment: Moving towards operational products. *Remote Sensing of Environment* 136, 455–468. doi:10.1016/j.rse.2013.05.029.

Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brock-
1125 mann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv,
M., Herold, M., Tsendbazar, N.E., Xu, P., Ramoino, F., Arino, O., 2022.
ESA WorldCover 10 m 2021 v200. Zenodo. doi:10.5281/zenodo.7254221.